

Troubleshooting Guide

Common Issues

1. UE5 Build Issues

“No GPU detected”

- **Cause:** UE5 requires a GPU for rendering and Pixel Streaming
- **Fix:** Use a GPU-enabled machine (AWS g4dn/g5, GCP N1+T4, Azure NV series)
- **Workaround:** Build scripts and asset prep can run on CPU-only machines

“GitHub 403 when cloning UE5”

- **Cause:** GitHub account not linked to Epic Games
- **Fix:**
 1. Go to <https://www.unrealengine.com/account/connections>
 2. Link your GitHub account
 3. Accept the Epic Games organization invite
 4. Use a Personal Access Token:

```
git clone https://<PAT>@github.com/EpicGames/UnrealEngine.git
```

Build fails with memory errors

- **Fix:** Ensure 32GB+ RAM; use `-MaxParallelActions 4` flag to reduce parallelism
-

2. Streaming Issues

WebSocket connection refused

```
# Check if signalling server is running
curl http://localhost:8888/healthz

# If not running:
cd streaming/signalling && npm start

# Check firewall
sudo ufw allow 8888
sudo ufw allow 443
```

Video stream not appearing

- Ensure UE5 server is running with `-PixelStreamingURL=ws://signalling:8888`
- Check browser console for WebRTC errors
- Verify TURN server is accessible: `turnutils_uclient turn.finalevolutiongroup.com`

High latency (>100ms)

- Move GPU server closer to users (same region)
- Check network congestion: `mtr <server_ip>`
- Reduce streaming resolution in UE5 config
- Enable hardware encoding (NVENC)

Black screen on mobile

- Check if H.264 is supported (iOS requires H.264)
 - Verify WebRTC codec negotiation in browser console
 - Test with VP8 fallback
-

3. iOS App Issues

Xcode signing errors

```
# Verify provisioning profile
security find-identity -v -p codesigning

# Clean build folder
xcodebuild clean -workspace FinalEvolutionLab.xcworkspace -scheme FinalEvolutionLab
```

App crashes on launch

- Check `Config.swift` URLs are correct
- Verify minimum iOS version is 16.0
- Check for missing privacy descriptions in `Info.plist`

Cannot connect to streaming server

- Verify ATS exceptions in `Info.plist`
 - Check if `stream.finalevolutiongroup.com` resolves on device
 - Test WebSocket connection: open Safari → `wss://stream.finalevolutiongroup.com`
-

4. DNS Issues

Domain not resolving

```
# Check propagation
dig @8.8.8.8 finalevolutiongroup.com
dig @1.1.1.1 stream.finalevolutiongroup.com

# Check nameservers
dig NS finalevolutiongroup.com

# Flush local DNS cache
sudo systemd-resolve --flush-caches # Linux
sudo dscacheutil -flushcache        # macOS
```

SSL certificate errors

- Verify cert covers all subdomains
 - Check Cloudflare SSL mode matches server config
 - Ensure stream/turn subdomains are not proxied through Cloudflare
-

5. Docker Issues

Container won't start

```
# Check logs
docker-compose logs -f signalling
docker-compose logs -f coturn

# Rebuild
docker-compose build --no-cache
docker-compose up -d
```

Port conflicts

```
# Find process using port
lsof -i :8888
lsof -i :3478

# Kill process
kill $(lsof -ti:8888)
```

6. Asset Pipeline Issues

DeepMotion API errors

- Verify `DEEPMOTION_CLIENT_ID` and `DEEPMOTION_CLIENT_SECRET` in `.env`
- Check API quota at <https://www.deepmotion.com/dashboard>
- Run `python scripts/process_elijahbonds_animations.py --test-auth`

Meshy generation failures

- Verify `MESHY_API_KEY` in `.env`
- Check task status: `python -c "from scripts.ai_asset_pipeline.meshy_service import MeshyService; ..."`
- API rate limit: wait 60s between requests

Missing assets in UE5

- Run `python scripts/ue5_setup/fel_verify_assets_standalone.py`
 - Check `GeneratedAssets/pipeline_report.json` for failures
 - Re-run import: `python UnrealStarter/BasketballGame/EditorPython/fel_import_ai_assets.py`
-

Emergency Procedures

Service Recovery

```
# Restart all services
./scripts/stop_services.sh
./scripts/start_services.sh

# Docker recovery
docker-compose down
docker-compose up -d

# Check health
curl http://localhost:8888/healthz
```

Rollback

```
# Git rollback to last known good state
git log --oneline -10
git checkout <commit_hash>

# Rebuild
npm run build # frontend
./scripts/start_services.sh
```

CV Preprocessing

“ultralalytics not installed” Warning

The pipeline falls back to OpenCV MOG2 background subtraction. For best results:

```
pip install ultralytics
# GPU acceleration (recommended):
pip install torch torchvision
```

Low Detection Rate (<70%)

- Try lowering confidence threshold: `--confidence 0.25`
- Video may have heavy occlusion or scene cuts
- Consider using SAM: `--use-sam` (requires GPU + `segment-anything`)
- Check if video has overlays/watermarks affecting detection

Identity Swap (Tracker Follows Wrong Person)

- The IoU tracker may swap to another player when actors cross paths
- Use SAM for pixel-precise masks: `--use-sam`
- Check validation report for “identity_swap_risk” warnings
- Manual review recommended for videos with >5 identity swap warnings

Hoop Not Detected

- Hoop detection works best with outdoor/clear courts
- Indoor footage with poor lighting may fail

- Ensure video shows hoop in upper portion of frame
- Try adjusting `--hoop-sample-rate 3` for more sampling

“No module named ‘cv2’” Error

```
pip install opencv-python-headless
```

Kalman Filter Producing Drift

- Increase measurement noise: adjust `measurement_noise` parameter
- Try `--method savgol` for non-predictive smoothing
- Use `--method combined` for best of both

Out of Memory During Processing

- Process videos one at a time: `--video <filename>`
- Reduce video resolution before processing
- Use `--no-preview` to skip preview video generation
- On GPU: ensure CUDA memory is not consumed by other processes

Validation Quality Score Too Low

- Review specific failed checks in the validation report
- Common fixes:
 - Lower confidence for better detection rate
 - Use SAM for better segmentation
 - Increase smoothing for jitter reduction
 - Re-process problematic videos individually

AWS GPU Deployment

G5 Instance Quota Error

```
Error: You have requested more vCPU capacity than your current vCPU limit
```

Request quota increase in AWS Console → Service Quotas → EC2 → Running On-Demand G and VT instances.

SSH Connection Timeout

- Security group may not have port 22 open
- Instance may still be initializing (wait 5 minutes)
- Check instance state: `python3 scripts/aws_deployment/provision_g5_instance.py --status`

NVIDIA Driver Not Loading

```
# Check driver status
nvidia-smi
# If not found, check setup log
cat /var/log/fel-setup.log
# May need reboot after driver install
sudo reboot
```

High AWS Costs

- **Always terminate instances** when not in use
- Use spot instances for 60-70% savings
- Monitor with: `python3 scripts/aws_deployment/provision_g5_instance.py --status`